

Empowering Cataloging Through AI : Practical Solutions and Explorations from the University of Miami Libraries

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Introduction

Artificial Intelligence (AI) is rapidly transforming how libraries approach technical services.

At the University of Miami Libraries (UML), AI is being explored not as a replacement for catalogers, but as a tool that can augment expertise, accelerate repetitive tasks, and enhance metadata quality.



Key Areas of Exploration



Metadata Creation & Enrichment

Exploring how AI tools can support the creation of new bibliographic and authority records:

- AI Assistants for Record Creation: Using AI to draft initial records based on book data, tables of contents, or abstracts.
- Automated Classification Suggestions: Leveraging AI to recommend Library of Congress (LCC) or Dewey Decimal (DDC) classifications based on content analysis.
- Content Enrichment: Generating summaries, proposing subject headings, and identifying relevant classification numbers to enhance metadata quality and discovery.



Data Remediation & Cleanup

Harnessing AI for large-scale metadata quality control and record consistency:

- AI-Assisted Normalization: Automatically standardizing fields such as author names, publication data, and subject terms.
- Error Detection & Correction: Identifying missing, inconsistent, or obsolete data and suggesting fixes, reducing the need for manual review.
- Pattern Recognition: Using machine learning to detect systemic issues in legacy records and batch-correct them.



Text Extraction & Analysis

Applying AI to interpret and analyze textual content from digital and physical sources.

- Optical Character Recognition (OCR): Extracting readable text from scanned materials and images.
- Semantic Analysis: Employing AI to detect subjects, names, and relationships within extracted text for automated cataloging.
- Classification Aids: Suggesting relevant Library of Congress Control Numbers (LCCNs) or subject headings based on textual patterns.



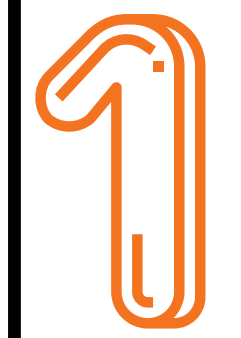
Translation & Transliteration

Enhancing multilingual discovery and accessibility through AI-driven linguistic tools..

- Translating bibliographic information, abstracts, and notes across languages while preserving context.
- Transliteration Tools: Converting scripts (e.g., Cyrillic, Arabic, Chinese) for consistent indexing and retrieval.

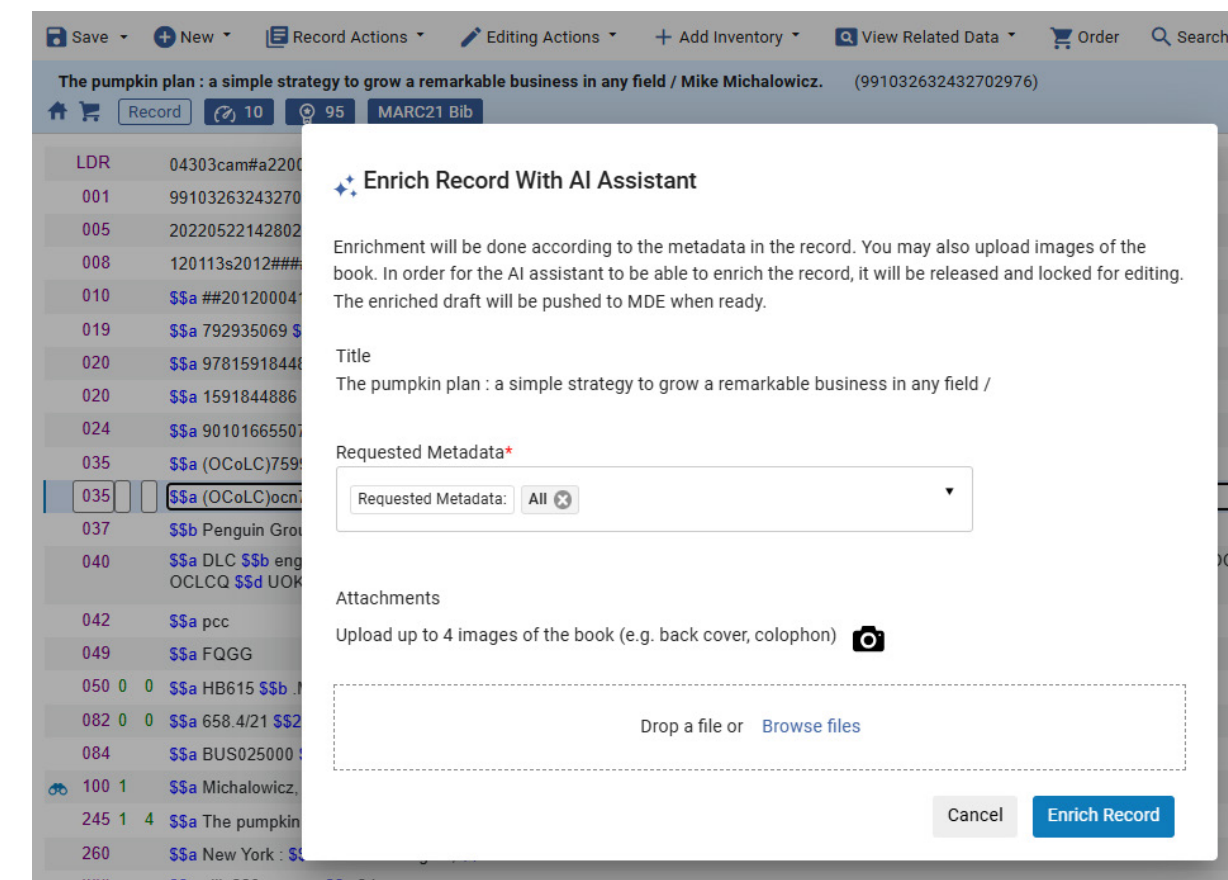
Examples from UML Initiatives

Metadata Generation



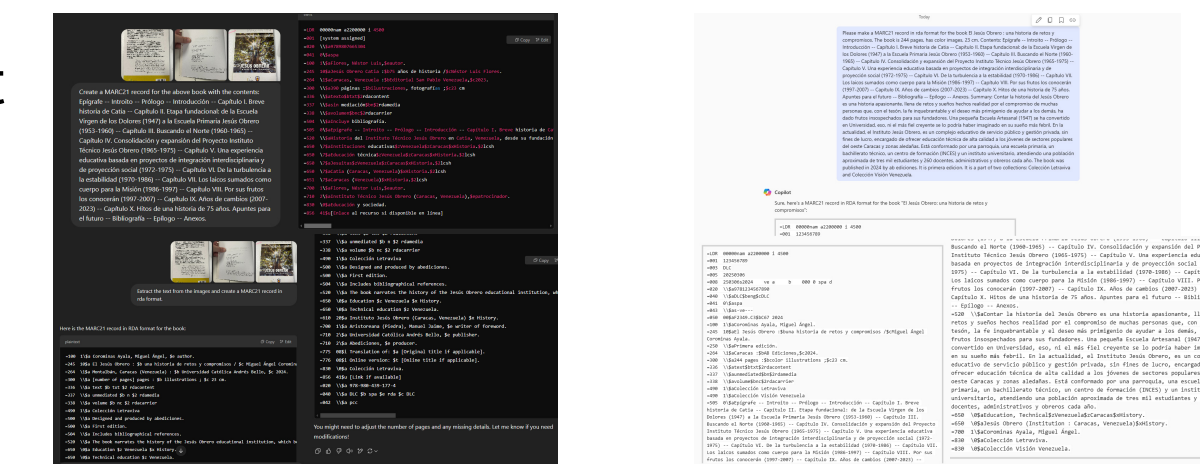
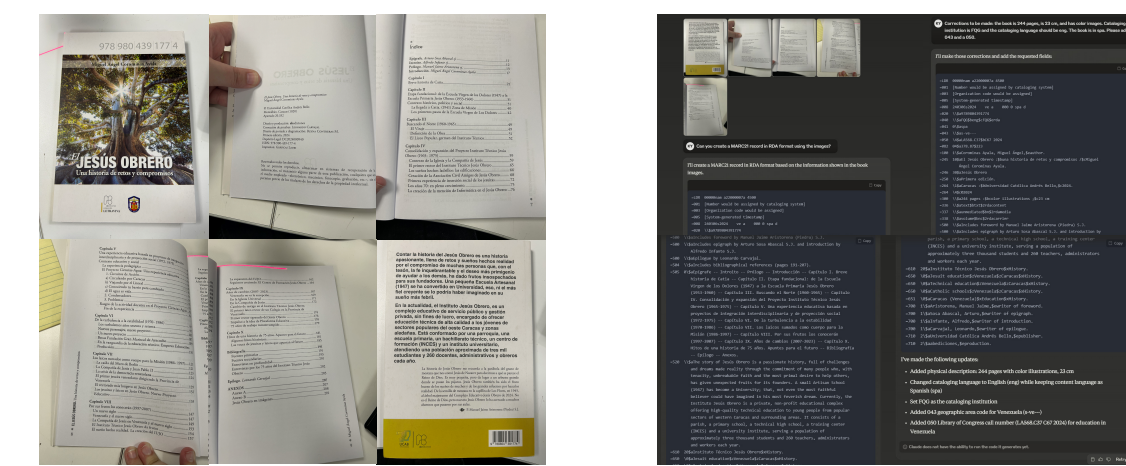
AI Metadata Assistant in Alma

- The AI Metadata Assistant uses a Large Language Model generative AI to process information about a library resource, and suggest relevant metadata to the cataloger.
- It can be used for creating new bibliographic records, as well as enriching existing brief records.
- The subjects provided are validated against Library of Congress vocabularies, with plans to increase the selection of authority vocabularies in future phases.

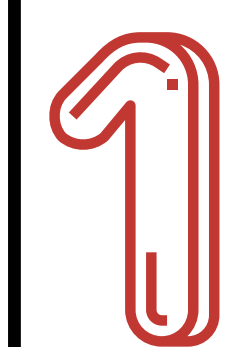


Comparison: ChatGPT, Copilot, Claude

- March 2025 - Comparison of Claude, Copilot, and ChatGPT
 - Claude needed less direction to produce a record
 - All of the AIs had trouble creating a call number and appropriate subject headings
- August 2025 - redid the comparison of the AIs.
 - ChatGPT-5 needed less direction than Claude 4 and Copilot and was able to suggest subject headings that were almost the same as LCSH



Data Remediation and Cleanup



AI-Assisted Workflow for Standardizing Item Descriptions

Implemented a workflow that integrates Alma, Excel, and ChatGPT for batch standardization of item descriptions:

- Data Extraction: Exported item records from Alma into an Excel spreadsheet, focusing on the Item Description field.
- Uploaded the spreadsheet to ChatGPT, prompting the model to identify and correct inconsistencies in formatting, punctuation, and descriptive patterns
- Reviewed AI-generated corrections for accuracy.
- Re-uploaded the standardized data into Alma using the Item Updater by Excel Cloud App, completing the



Pattern Recognition:

Upload a MARC file to ChatGPT to find errors, inconsistencies/cleanup for normalization, and suggested improvements.

Record 6 — 991009925769702976 (Beyond slavery ..., 2007)

Errors / high-priority

- Duplicated, conflicting 505 fields: you have two 505 0_ with overlapping but non-identical contents. Keep one accurate 505, or merge into a single structured 505 (e.g., \$t /\$r with part labels as you've done elsewhere).

Record 3 — 991031334787402976 (Afro-Asian connections ..., 2019)

- Errors / high-priority
- MARC/RDA coding omissions in content/carrier set:
 - 336 missing \$b txt (should be \$a text \$b txt \$2 rdacontent).
 - 337 missing \$b n (should be \$a mediated \$b n \$2 rdaMedia).
 - 338 missing \$b nc (should be \$a volume \$b nc \$2 rdaCarrier).
 - Punctuation in 300 : illustrations : \$c 24 cm → second colon should be a semicolon (;)
 - Illustrations : \$c 24 cm
 - Obsolete pre-pub field. 363 \$a 1981 (projected pub date) should be removed post-publication.
 - Duplicate locals: multiple 852 and 982.

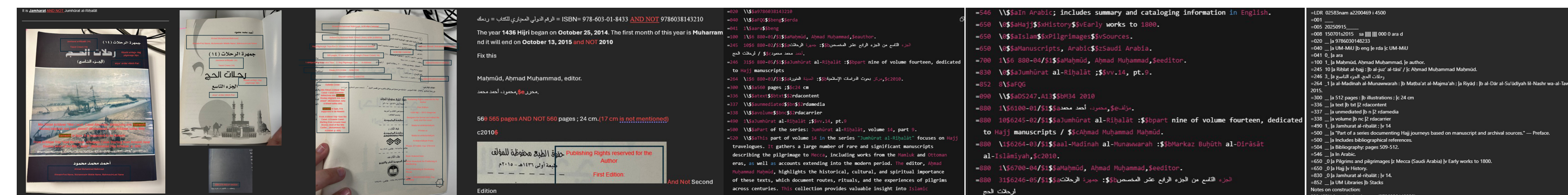
Inconsistencies / cleanup

- Series:
 - 490 1_ \$a Black diasporic worlds: origins, and evolutions from new world slavery vs 490 2_ \$a Black diasporic worlds → generally fine (490 uses authorized form, 490 is transcribed), just

Translation and Transliteration

AI-Generated Transliterated Record

- Created a record for a book in Arabic using the free version of ChatGPT-4.
- Text extraction played the biggest role in this experiment. ChatGPT uses OCR to pull information from the uploaded images.
- Layal Hneiny, UM's Clinical Research Librarian, reviewed the free version record. The title, the ISBN, the page count, the summary, and copyright date were all wrong. Pictured below are all her translations and notes.
- Ran it through the paid version and compared it to Layal's notes; this time the title, copyright date, and edition statement were right.



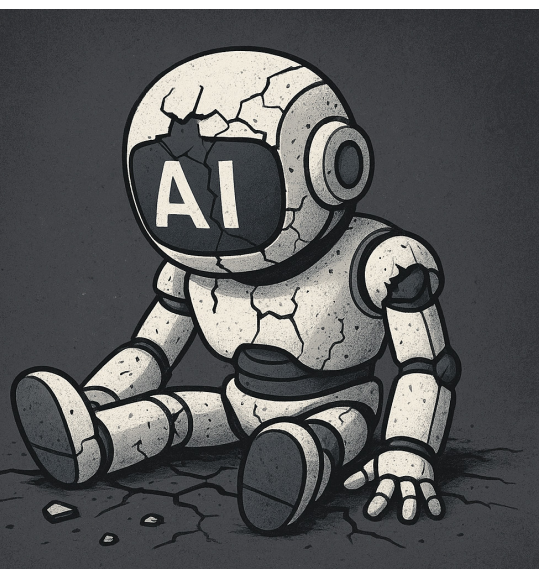
Generation of summaries for Marine Theses

- Objective: Generate concise AI-based summaries for over 1,000 RSMAS theses to improve discovery and access.
- Method: Each thesis is represented by three scanned images named with its unique 14-digit barcode (e.g., 35051104044120_1.jpg, _2, _3).
- Process: ChatGPT analyzes the images and produces 1–2 paragraph summaries compiled in Excel (barcode + summary).

Barcode	MMS id	Summary
35051104044120	99100382629702976	Michael Charles Schmidt's 1979 master's thesis in Marine Biology examines sexual selection and reproductive success in males of the bicolor damselfish (Eupomacentrus partitus). The study investigates how contribute to reproductive success in a territorial Caribbean species. Over 14 months, male guarding demersal eggs within territories showed significant variation in spawning success. The research found sex and frequency of courtship, aggression, and male size. Non-random mating patterns highlighted the selective importance of these traits.
35051104044120	99100382629702976	The introduction places the study in the context of Darwin's theory of sexual selection, including female mate choice and Fisher's (1930) model of self-accelerating selection. The work explores the relationship competition, and female choice, with references to broader evolutionary and ecological theory. Results suggest that female mate choice is the dominant driver of sexual selection in this species, shaping male contributing to the broader understanding of sexual selection in fishes.
35051104044120	99100382629702976	Ping-Hong Chen's 1996 master's thesis in Meteorology and Physical Oceanography investigates wind forced geostrophic transport in the Straits of Florida using numerical simulations. The study focuses on if stratification changes in response to modified surface wind stress. Using circulation models with controlled inflow conditions, Chen examines how varying wind directions affect surface currents, revealing per reaching the Florida coast.
35051104044120	99100382629702976	The research introduces the concept of C-stress (current-induced stress) and compares it with traditional wind stress models. Results show that neglecting ocean currents in stress calculations leads to substrate of Florida, where wind and currents interact strongly. Findings reveal that the structure of surface stress patterns is anti-correlated with surface currents and that C-stress significantly influences mesoscale or practical implications for understanding marine ecology, coastal water quality, and the movement of boats and pollutants.

Challenges and Possibilities

- Bias
 - Large language models (LLMs) may reflect biases or inaccuracies present in their training data.
- Quality
 - Takes things out of context and runs with them
 - Cannot reliably identify or apply authorized names or subject headings
- Access and cost:
 - Paid version of ChatGPT provides better "thought processes" and works faster that the free version as shown by the the transliteration experiment.
 - Not all Technical Services staff have access to paid AI tool subscriptions.
- Language limitations: Seems to perform best with English titles
- Processing constraints: ChatGPT Premium only allows us to upload up to 10 images at the same time, which meant that we could only generate summaries for 3 theses at a time (3 images per thesis). That is not ideal since we want to produce summaries for more than 1,000 theses.



Conclusions

- There is significant opportunity for further exploration of AI in cataloging.
- Human expertise remains essential for ensuring metadata quality. AI should complement, not replace, catalogers' judgment.
- We are currently exploring AI tools through small-scale projects, but they have not yet been implemented in regular workflows. As AI continues to evolve and its results remain imperfect, we are approaching its use with caution and patience—prioritizing accuracy and reliability over speed.



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